

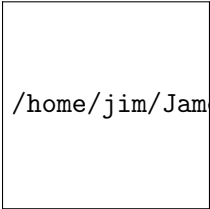
Section 3.4: Combinatorics & Probability

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Monday 13th October, 2014 & Wednesday 15th October, 2014



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Combinatorics review/ 3 types of Questions

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Example 3.4.1.

How many ways can you roll two 6 sided dice?

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- can repeat
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$$6 \times 6$$

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Combination

$${}_nC_r = \frac{n!}{(n-r)!r!}$$

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$${}_{40}C_5$$

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$${}_nC_r = \frac{n!}{(n-r)!r!}$$

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$${}_nC_r = \frac{n!}{(n-r)!r!}$$

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What is the probability that someone rolls a 3 then a 2?

$$p(3 \text{ and } 2) = \frac{n(3 \text{ and } 2)}{n(S)} = \frac{1}{36}$$

Combinatorics & Probability: 3 types of Questions

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E' : 3 people have different birthdays.

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$$p(E') = \frac{n(E')}{n(S)}$$

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$$= 0.008204\dots$$

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$$= 0.008204\dots$$

$$\approx 0.8\%$$

Combinatorics & Probability: 3 types of Questions

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What is the probability that someone matches 5 randomly drawn numbers from 1-40?

- A 5/40 lottery game.

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How many ways can you select 5 numbers 1-40?

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ex. If $\{10, 11, 20, 40, 1\}$ is the drawn
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$$p(E) = \frac{n(E)}{n(S)}$$

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$$\begin{aligned} p(E) &= \frac{n(E)}{n(S)} = \frac{1}{40 C_5} \\ &= \frac{1}{(40 \cdot 39 \cdot 38 \cdot 37 \cdot 36 / 120)} \\ &= \frac{1}{7,059,052} \end{aligned}$$

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$$\begin{aligned} p(E) &= \frac{n(E)}{n(S)} = \frac{1}{40 C_5} \\ &= \frac{1}{(40 \cdot 39 \cdot 38 \cdot 37 \cdot 36 / 120)} \\ &= \frac{1}{7,059,052} \\ &\approx 0.00001417\% \end{aligned}$$

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What is the probability that at least two of them have the same birthday?

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What is the probability that at least two of them have the same birthday?

$$n(S) = 365^{30} \approx 7.39241 \cdot 10^{76} \text{ (very big)}$$

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$$n(E') = {}_{365}P_{30} \approx 2.171103 \cdot 10^{76} \text{ (also very big)}$$

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$$\begin{aligned} p(E) &= 1 - p(E') = 1 - \frac{n(E')}{n(S)} \\ &= 1 - \frac{{}_{365}P_{30}}{365^{30}} \end{aligned}$$

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Problem 3.4.2. A group of 35 people is selected at random

What is the probability that at least two of them have the same birthday?

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What is the probability that at least two of them have the same birthday?

$$n(S) = 365^{35}$$

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What is the probability that at least two of them have the same birthday?

$$n(S) = 365^{35}$$

Let E : At least two have the same birthday.

Problem 3.4.2. A group of 35 people is selected at random

What is the probability that at least two of them have the same birthday?

$$n(S) = 365^{35}$$

Let E : At least two have the same birthday.

$$n(E') =$$

Problem 3.4.2. A group of 35 people is selected at random

What is the probability that at least two of them have the same birthday?

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Let E : At least two have the same birthday.

$$n(E') = {}_{365}P_{35}$$

Problem 3.4.2. A group of 35 people is selected at random

What is the probability that at least two of them have the same birthday?

$$n(S) = 365^{35}$$

Let E : At least two have the same birthday.

$$n(E') = {}_{365}P_{35}$$

$$p(E) =$$

Problem 3.4.2. A group of 35 people is selected at random

What is the probability that at least two of them have the same birthday?

$$n(S) = 365^{35}$$

Let E : At least two have the same birthday.

$$n(E') = {}_{365}P_{35}$$

$$\begin{aligned} p(E) &= 1 - p(E') = 1 - \frac{n(E')}{n(S)} \\ &= 1 - \frac{{}_{365}P_{35}}{365^{35}} \end{aligned}$$

Problem 3.4.2. A group of 35 people is selected at random

What is the probability that at least two of them have the same birthday?

$$n(S) = 365^{35}$$

Let E : At least two have the same birthday.

$$n(E') = {}_{365}P_{35}$$

$$\begin{aligned} p(E) &= 1 - p(E') = 1 - \frac{n(E')}{n(S)} \\ &= 1 - \frac{{}_{365}P_{35}}{365^{35}} \\ &\approx .8144 \end{aligned}$$

Problem 3.4.3.

What is the least number you would need in a group so that there is a probability of *at least* 78%?

From the last two problems we know the answer is between 30 and 35

Problems

From the last two problems we know the answer is between 30 and 35

$$p(E_{30}) = 1 - p(E'_{30}) = 1 - \frac{{}^{365}P_{30}}{365^{30}} \approx .7063$$

Problems

From the last two problems we know the answer is between 30 and 35

$$p(E_{30}) = 1 - p(E'_{30}) = 1 - \frac{{}^{365}P_{30}}{365^{30}} \approx .7063$$

$$p(E_{35}) = 1 - p(E'_{35}) = 1 - \frac{{}^{365}P_{35}}{365^{35}} \approx .8144$$

Problems

From the last two problems we know the answer is between 30 and 35

$$p(E_{30}) = 1 - p(E'_{30}) = 1 - \frac{{}^{365}P_{30}}{365^{30}} \approx .7063$$

$$p(E_{31}) = 1 - p(E'_{31}) = 1 - \frac{{}^{365}P_{31}}{365^{31}} \approx .7305$$

$$p(E_{35}) = 1 - p(E'_{35}) = 1 - \frac{{}^{365}P_{35}}{365^{35}} \approx .8144$$

From the last two problems we know the answer is between 30 and 35

$$p(E_{30}) = 1 - p(E'_{30}) = 1 - \frac{{}^{365}P_{30}}{365^{30}} \approx .7063$$

$$p(E_{31}) = 1 - p(E'_{31}) = 1 - \frac{{}^{365}P_{31}}{365^{31}} \approx .7305$$

$$p(E_{32}) = 1 - p(E'_{32}) = 1 - \frac{{}^{365}P_{32}}{365^{32}} \approx .7533$$

$$p(E_{35}) = 1 - p(E'_{35}) = 1 - \frac{{}^{365}P_{35}}{365^{35}} \approx .8144$$

From the last two problems we know the answer is between 30 and 35

$$p(E_{30}) = 1 - p(E'_{30}) = 1 - \frac{{}^{365}P_{30}}{365^{30}} \approx .7063$$

$$p(E_{31}) = 1 - p(E'_{31}) = 1 - \frac{{}^{365}P_{31}}{365^{31}} \approx .7305$$

$$p(E_{32}) = 1 - p(E'_{32}) = 1 - \frac{{}^{365}P_{32}}{365^{32}} \approx .7533$$

$$p(E_{33}) = 1 - p(E'_{33}) = 1 - \frac{{}^{365}P_{33}}{365^{33}} \approx .7749$$

$$p(E_{35}) = 1 - p(E'_{35}) = 1 - \frac{{}^{365}P_{35}}{365^{35}} \approx .8144$$

From the last two problems we know the answer is between 30 and 35

$$p(E_{30}) = 1 - p(E'_{30}) = 1 - \frac{{}^{365}P_{30}}{365^{30}} \approx .7063$$

$$p(E_{31}) = 1 - p(E'_{31}) = 1 - \frac{{}^{365}P_{31}}{365^{31}} \approx .7305$$

$$p(E_{32}) = 1 - p(E'_{32}) = 1 - \frac{{}^{365}P_{32}}{365^{32}} \approx .7533$$

$$p(E_{33}) = 1 - p(E'_{33}) = 1 - \frac{{}^{365}P_{33}}{365^{33}} \approx .7749$$

$$p(E_{34}) = 1 - p(E'_{34}) = 1 - \frac{{}^{365}P_{34}}{365^{34}} \approx .7953$$

$$p(E_{35}) = 1 - p(E'_{35}) = 1 - \frac{{}^{365}P_{35}}{365^{35}} \approx .8144$$

Problems

From the last two problems we know the answer is between 30 and 35

$$p(E_{30}) = 1 - p(E'_{30}) = 1 - \frac{{}^{365}P_{30}}{365^{30}} \approx .7063$$

$$p(E_{31}) = 1 - p(E'_{31}) = 1 - \frac{{}^{365}P_{31}}{365^{31}} \approx .7305$$

$$p(E_{32}) = 1 - p(E'_{32}) = 1 - \frac{{}^{365}P_{32}}{365^{32}} \approx .7533$$

$$p(E_{33}) = 1 - p(E'_{33}) = 1 - \frac{{}^{365}P_{33}}{365^{33}} \approx .7749$$

$$p(E_{34}) = 1 - p(E'_{34}) = 1 - \frac{{}^{365}P_{34}}{365^{34}} \approx .7953 \leftarrow \text{above } 78\%$$

$$p(E_{35}) = 1 - p(E'_{35}) = 1 - \frac{{}^{365}P_{35}}{365^{35}} \approx .8144$$

Problem 3.4.4. A state lottery has a 5/40 game (pick 5 different numbers from 1-40).

$$\begin{aligned} \text{a. } p(1^{\text{st}}) &= \frac{n(1^{\text{st}})}{n(S)} \\ &= \frac{{}_5C_5}{{}_{40}C_5} \leftarrow \begin{array}{l} \text{ways to choose 5 of 5 winning numbers} \\ \text{ways to choose 40 of 5 numbers} \end{array} \\ &= \frac{1}{658008} \approx 0.000002 \end{aligned}$$

- a. $p(1^{\text{st}}) = \frac{n(1^{\text{st}})}{n(S)}$
 $= \frac{{}_5C_5}{{}_{40}C_5} \leftarrow \text{ways to choose 5 of 5 winning numbers}$
 $= \frac{1}{658008} \approx 0.000002$
- b. $n(2^{\text{nd}}) = (\text{ways to pick 4 winners}) \times (\text{ways to choose 1 loser})$
 $= {}_5C_4 \times {}_{35}C_1 = 5 \times 35 = 175$
 $p(2^{\text{nd}}) = \frac{n(2^{\text{nd}})}{n(S)} = \frac{175}{658008} \approx .000266$

- a. $p(1^{\text{st}}) = \frac{n(1^{\text{st}})}{n(S)}$
 $= \frac{{}_5C_5}{{}_{40}C_5} \leftarrow \begin{array}{l} \text{ways to choose 5 of 5 winning numbers} \\ \text{ways to choose 40 of 5 numbers} \end{array}$
 $= \frac{1}{658008} \approx 0.000002$
- b. $n(2^{\text{nd}}) = (\text{ways to pick 4 winners}) \times (\text{ways to choose 1 loser})$
 $= {}_5C_4 \times {}_{35}C_1 = 5 \times 35 = 175$
 $p(2^{\text{nd}}) = \frac{n(2^{\text{nd}})}{n(S)} = \frac{175}{658008} \approx .000266$
- Note: ${}_{35}C_1 = 35 = (40 - 5)$, the non-winning numbers.

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

Find the probability of being dealt 2 aces, 2 queens, and a king?

Problems

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

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Total number of hands, $n(S) =$

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

Find the probability of being dealt 2 aces, 2 queens, and a king?

Total number of hands, $n(S) = {}_{52}C_5$ (order doesn't matter)

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

Find the probability of being dealt 2 aces, 2 queens, and a king?

Total number of hands, $n(S) = {}_{52}C_5$ (order doesn't matter)

Number of ways to get 2 aces, $n(2A) =$

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

Find the probability of being dealt 2 aces, 2 queens, and a king?

Total number of hands, $n(S) = {}_{52}C_5$ (order doesn't matter)

Number of ways to get 2 aces, $n(2A) = {}_4C_2$

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

Find the probability of being dealt 2 aces, 2 queens, and a king?

Total number of hands, $n(S) = {}_{52}C_5$ (order doesn't matter)

Number of ways to get 2 aces, $n(2A) = {}_4C_2$

Number of ways to get 2 queens, $n(2Q) =$

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

Find the probability of being dealt 2 aces, 2 queens, and a king?

Total number of hands, $n(S) = {}_{52}C_5$ (order doesn't matter)

Number of ways to get 2 aces, $n(2A) = {}_4C_2$

Number of ways to get 2 queens, $n(2Q) = {}_4C_2$

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

Find the probability of being dealt 2 aces, 2 queens, and a king?

Total number of hands, $n(S) = {}_{52}C_5$ (order doesn't matter)

Number of ways to get 2 aces, $n(2A) = {}_4C_2$

Number of ways to get 2 queens, $n(2Q) = {}_4C_2$

Number of ways to get a king, $n(1K) =$

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

Find the probability of being dealt 2 aces, 2 queens, and a king?

Total number of hands, $n(S) = {}_{52}C_5$ (order doesn't matter)

Number of ways to get 2 aces, $n(2A) = {}_4C_2$

Number of ways to get 2 queens, $n(2Q) = {}_4C_2$

Number of ways to get a king, $n(1K) = {}_4C_1$

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

Find the probability of being dealt 2 aces, 2 queens, and a king?

Total number of hands, $n(S) = {}_{52}C_5$ (order doesn't matter)

Number of ways to get 2 aces, $n(2A) = {}_4C_2$

Number of ways to get 2 queens, $n(2Q) = {}_4C_2$

Number of ways to get a king, $n(1K) = {}_4C_1$

$$p(2A \cap 2Q \cap 1K) = \frac{n(2A \cap 2Q \cap 1K)}{n(S)} =$$

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

Find the probability of being dealt 2 aces, 2 queens, and a king?

Total number of hands, $n(S) = {}_{52}C_5$ (order doesn't matter)

Number of ways to get 2 aces, $n(2A) = {}_4C_2$

Number of ways to get 2 queens, $n(2Q) = {}_4C_2$

Number of ways to get a king, $n(1K) = {}_4C_1$

$$p(2A \cap 2Q \cap 1K) = \frac{n(2A \cap 2Q \cap 1K)}{n(S)} = \frac{n(2A) \times n(2Q) \times n(1K)}{n(S)}$$

Problem 3.4.5. 5 cards are dealt (without replacement) from a standard 52 card deck.

Find the probability of being dealt 2 aces, 2 queens, and a king?

Total number of hands, $n(S) = {}_{52}C_5$ (order doesn't matter)

Number of ways to get 2 aces, $n(2A) = {}_4C_2$

Number of ways to get 2 queens, $n(2Q) = {}_4C_2$

Number of ways to get a king, $n(1K) = {}_4C_1$

$$\begin{aligned} p(2A \cap 2Q \cap 1K) &= \frac{n(2A \cap 2Q \cap 1K)}{n(S)} = \frac{n(2A) \times n(2Q) \times n(1K)}{n(S)} \\ &= \frac{{}_4C_2 \times {}_4C_2 \times {}_4C_1}{{}_{52}C_5} = \frac{6 \times 6 \times 4}{2598960} = \frac{3}{54145} \end{aligned}$$

Problems

Problem 3.4.6. You order 12 burritos to go, 5 with peppers and 7 without peppers. However the restaurant did not label the burritos. If you pick 3 burritos at random find the probability of each of the follow:

None have peppers.

Problem 3.4.6. You order 12 burritos to go, 5 with peppers and 7 without peppers. However the restaurant did not label the burritos. If you pick 3 burritos at random find the probability of each of the follow:

None have peppers.

Need to find:

$$p(NP) = \frac{n(NP)}{n(S)} = \frac{\text{ways to pick 3 w/no peppers}}{\text{ways to pick 3 burritos}}$$

Problem 3.4.6. You order 12 burritos to go, 5 with peppers and 7 without peppers. However the restaurant did not label the burritos. If you pick 3 burritos at random find the probability of each of the follow:

None have peppers.

Need to find:

$$p(NP) = \frac{n(NP)}{n(S)} = \frac{\text{ways to pick 3 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) =$$

Problem 3.4.6. You order 12 burritos to go, 5 with peppers and 7 without peppers. However the restaurant did not label the burritos. If you pick 3 burritos at random find the probability of each of the follow:

None have peppers.

Need to find:

$$p(NP) = \frac{n(NP)}{n(S)} = \frac{\text{ways to pick 3 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

Problem 3.4.6. You order 12 burritos to go, 5 with peppers and 7 without peppers. However the restaurant did not label the burritos. If you pick 3 burritos at random find the probability of each of the follow:

None have peppers.

Need to find:

$$p(NP) = \frac{n(NP)}{n(S)} = \frac{\text{ways to pick 3 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$n(NP) =$$

Problem 3.4.6. You order 12 burritos to go, 5 with peppers and 7 without peppers. However the restaurant did not label the burritos. If you pick 3 burritos at random find the probability of each of the follow:

None have peppers.

Need to find:

$$p(NP) = \frac{n(NP)}{n(S)} = \frac{\text{ways to pick 3 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$n(NP) = {}_7C_3 = 35$$

Problem 3.4.6. You order 12 burritos to go, 5 with peppers and 7 without peppers. However the restaurant did not label the burritos. If you pick 3 burritos at random find the probability of each of the follow:

None have peppers.

Need to find:

$$p(NP) = \frac{n(NP)}{n(S)} = \frac{\text{ways to pick 3 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$n(NP) = {}_7C_3 = 35$$

$$p(NP) = \frac{{}_7C_3}{{}_{12}C_3} = \frac{35}{220} = \frac{7}{44}$$

Problem 3.4.7. You order 12 burritos to go, 5 with peppers and 7 without peppers. If you pick 3 burritos at random find the probability of each of the follow:

Exactly 2 have no peppers.

Need to find:

$$p(NP) = \frac{n(2P)}{n(S)} = \frac{\text{ways to pick 2 w/no peppers}}{\text{ways to pick 3 burritos}}$$

Problems

Need to find:

$$p(NP) = \frac{n(2P)}{n(S)} = \frac{\text{ways to pick 2 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) =$$

Problems

Need to find:

$$p(NP) = \frac{n(2P)}{n(S)} = \frac{\text{ways to pick 2 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

Need to find:

$$p(NP) = \frac{n(2P)}{n(S)} = \frac{\text{ways to pick 2 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$n(2P) = (\text{ways to pick 2 w/no peppers}) \times (\text{ways to pick 1 w/peppers})$$

Need to find:

$$p(NP) = \frac{n(2P)}{n(S)} = \frac{\text{ways to pick 2 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$\begin{aligned} n(2P) &= (\text{ways to pick 2 w/no peppers}) \times (\text{ways to pick 1 w/peppers}) \\ &= {}_7C_2 \times {}_5C_1 = 21 \times 5 = 105 \end{aligned}$$

Need to find:

$$p(NP) = \frac{n(2P)}{n(S)} = \frac{\text{ways to pick 2 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$\begin{aligned} n(2P) &= (\text{ways to pick 2 w/no peppers}) \times (\text{ways to pick 1 w/peppers}) \\ &= {}_7C_2 \times {}_5C_1 = 21 \times 5 = 105 \end{aligned}$$

$$p(2P) = \frac{{}_7C_2 \times {}_5C_1}{{}_{12}C_3} = \frac{105}{220} = \frac{21}{44}$$

Problems

Problem 3.4.8. You order 12 burritos to go, 5 with peppers and 7 without peppers. However the restaurant did not label the burritos. If you pick 3 burritos at random find the probability of each of the follow:

At least 1 has no peppers.

Equivalently, 1 OR 2 OR 3 have peppers.

- At least, at most, more than, greater than, etc. → **or** → **add**

Equivalently, 1 OR 2 OR 3 have peppers.

- At least, at most, more than, greater than, etc. → **or** → **add**

Need to find:

$$p(NP) = \frac{n(NP \geq 1)}{n(S)} = \frac{\text{ways to get at least 1 w/no peppers}}{\text{ways to pick 3 burritos}}$$

Equivalently, 1 OR 2 OR 3 have peppers.

- At least, at most, more than, greater than, etc. → **or** → **add**

Need to find:

$$p(NP) = \frac{n(NP \geq 1)}{n(S)} = \frac{\text{ways to get at least 1 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) =$$

Equivalently, 1 OR 2 OR 3 have peppers.

- At least, at most, more than, greater than, etc. → **or** → **add**

Need to find:

$$p(NP) = \frac{n(NP \geq 1)}{n(S)} = \frac{\text{ways to get at least 1 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

Equivalently, 1 OR 2 OR 3 have peppers.

- At least, at most, more than, greater than, etc. → **or** → **add**

Need to find:

$$p(NP) = \frac{n(NP \geq 1)}{n(S)} = \frac{\text{ways to get at least 1 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$n(NP \geq 1) =$$

$$\begin{aligned} & n(1 \text{ w/no peppers}) \\ & + n(2 \text{ w/no peppers}) \\ & + n(3 \text{ w/no peppers}) \end{aligned}$$

Equivalently, 1 OR 2 OR 3 have peppers.

- At least, at most, more than, greater than, etc. $\rightarrow$ **or** $\rightarrow$ **add**

Need to find:

$$p(NP) = \frac{n(NP \geq 1)}{n(S)} = \frac{\text{ways to get at least 1 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$n(NP \geq 1) =$$

$$\begin{aligned} & n(1 \text{ w/no peppers}) \\ & + n(2 \text{ w/no peppers}) \\ & + n(3 \text{ w/no peppers}) \end{aligned}$$

$$= \quad + n(2 \text{ w/no peppers}) \times n(\text{pick 1 w/peppers}) \quad = \quad + {}_7C_2 \times {}_5C_1 \quad 105$$

Equivalently, 1 OR 2 OR 3 have peppers.

- At least, at most, more than, greater than, etc. → **or** → **add**

Need to find:

$$p(NP) = \frac{n(NP \geq 1)}{n(S)} = \frac{\text{ways to get at least 1 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$n(NP \geq 1) =$$

$$n(1 \text{ w/no peppers})$$

$$+ n(2 \text{ w/no peppers})$$

$$+ n(3 \text{ w/no peppers})$$

$$\begin{aligned} & n(1 \text{ w/no peppers}) \times n(\text{pick 2 w/peppers}) &= {}_7C_1 \times {}_5C_2 &= 70 \\ = & + n(2 \text{ w/no peppers}) \times n(\text{pick 1 w/peppers}) &= + {}_7C_2 \times {}_5C_1 &= 105 \end{aligned}$$

Equivalently, 1 OR 2 OR 3 have peppers.

- At least, at most, more than, greater than, etc. → **or** → **add**

Need to find:

$$p(NP) = \frac{n(NP \geq 1)}{n(S)} = \frac{\text{ways to get at least 1 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$n(NP \geq 1) =$$

$$n(1 \text{ w/no peppers})$$

$$+ n(2 \text{ w/no peppers})$$

$$+ n(3 \text{ w/no peppers})$$

$$\begin{aligned} & n(1 \text{ w/no peppers}) \times n(\text{pick 2 w/peppers}) &= & {}_7C_1 \times {}_5C_2 &= & 70 \\ = & + n(2 \text{ w/no peppers}) \times n(\text{pick 1 w/peppers}) &= & + {}_7C_2 \times {}_5C_1 &= & 105 \\ & + n(3 \text{ w/no peppers}) \times n(\text{pick 0 w/peppers}) &= & + {}_7C_3 \times {}_5C_0 &= & 35 \end{aligned}$$

Equivalently, 1 OR 2 OR 3 have peppers.

- At least, at most, more than, greater than, etc. → **or** → **add**

Need to find:

$$p(NP) = \frac{n(NP \geq 1)}{n(S)} = \frac{\text{ways to get at least 1 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$n(NP \geq 1) = 70 + 105 + 35$$

$n(1 \text{ w/no peppers})$

$+n(2 \text{ w/no peppers})$

$+n(3 \text{ w/no peppers})$

$$\begin{aligned} & n(1 \text{ w/no peppers}) \times n(\text{pick 2 w/peppers}) &= & {}_7C_1 \times {}_5C_2 &= & 70 \\ = & +n(2 \text{ w/no peppers}) \times n(\text{pick 1 w/peppers}) &= & +{}_7C_2 \times {}_5C_1 &= & 105 \\ & +n(3 \text{ w/no peppers}) \times n(\text{pick 0 w/peppers}) &= & +{}_7C_3 \times {}_5C_0 &= & 35 \end{aligned}$$

Equivalently, 1 OR 2 OR 3 have peppers.

- At least, at most, more than, greater than, etc. → **or** → **add**

Need to find:

$$p(NP) = \frac{n(NP \geq 1)}{n(S)} = \frac{\text{ways to get at least 1 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$n(NP \geq 1) = 70 + 105 + 35$$

$$n(1 \text{ w/no peppers})$$

$$+ n(2 \text{ w/no peppers})$$

$$+ n(3 \text{ w/no peppers})$$

$$\begin{aligned} & n(1 \text{ w/no peppers}) \times n(\text{pick 2 w/peppers}) \\ = & + n(2 \text{ w/no peppers}) \times n(\text{pick 1 w/peppers}) \\ & + n(3 \text{ w/no peppers}) \times n(\text{pick 0 w/peppers}) \end{aligned} \quad \begin{array}{r} {}_7C_1 \times {}_5C_2 \quad 70 \\ + {}_7C_2 \times {}_5C_1 \quad 105 \\ + {}_7C_3 \times {}_5C_0 \quad 35 \end{array}$$

$$p(NP \geq 1) = \frac{70 + 105 + 35}{220} = \frac{210}{220} = \frac{21}{22}$$

Equivalently, 1 OR 2 OR 3 have peppers.

- At least, at most, more than, greater than, etc. → **or** → **add**

Need to find:

$$p(NP) = \frac{n(NP \geq 1)}{n(S)} = \frac{\text{ways to get at least 1 w/no peppers}}{\text{ways to pick 3 burritos}}$$

$$n(S) = {}_{12}C_3 = 220$$

$$n(NP \geq 1) = 70 + 105 + 35$$

$$\begin{aligned} & n(1 \text{ w/no peppers}) \\ & + n(2 \text{ w/no peppers}) \\ & + n(3 \text{ w/no peppers}) \\ = & \begin{array}{ll} n(1 \text{ w/no peppers}) \times n(\text{pick 2 w/peppers}) & {}_7C_1 \times {}_5C_2 \quad 70 \\ + n(2 \text{ w/no peppers}) \times n(\text{pick 1 w/peppers}) & + {}_7C_2 \times {}_5C_1 \quad 105 \\ + n(3 \text{ w/no peppers}) \times n(\text{pick 0 w/peppers}) & + {}_7C_3 \times {}_5C_0 \quad 35 \end{array} \end{aligned}$$

$$p(NP \geq 1) = \frac{70 + 105 + 35}{220} = \frac{210}{220} = \frac{21}{22}$$

$$(1 - p(\text{all peppers}) = 1 - \frac{{}_5C_3}{220})$$

Online Resources

WolframAlpha

- www.wolframalpha.com

easycalculation.com

Online Resources

WolframAlpha

- www.wolframalpha.com

- ${}_nC_r = c(n, r)$

easycalculation.com

WolframAlpha

- www.wolframalpha.com

- ${}_nC_r = c(n, r)$

- ${}_nP_r = p(n, r)$

easycalculation.com

WolframAlpha

- www.wolframalpha.com
 - ${}_nC_r = c(n, r)$
 - ${}_nP_r = p(n, r)$

easycalculation.com

- www.easycalculation.com/statistics/permutation-combination.php